

JONATHAN KIM

(915) 474-8685 | El Paso, TX

jon.kim.mj@gmail.com | linkedin.com/in/jonathan-kim-j1k3 | github.com/jonkim13 | jonathan-kim.me

Education

The University of Texas at El Paso

Expected: May 2027

Bachelor of Science in Computer Science

GPA: 3.57/4.00

Relevant Coursework: Agentic AI, Machine Learning, Software Engineering, Data Structures & Algorithms, Database Systems, Operating Systems, Computer Organization, Computer Security

Activities: Secretary, Free and Open-Source Club (2024–2025)

Experience

Co-Founder & Frontend Engineer

December 2025 – Present

BorderETA | 1881 Ventures Accelerator | bordereta.live

El Paso, Texas

- Launched a bilingual iOS/Android app for commuters at 5 international border crossings, reaching 800+ users
- Engineered 10 nested geofences to replace battery-draining polling within iOS's 20-region cap
- Eliminated crossing data loss in cellular dead zones by implementing a write-ahead queue replayed on relaunch
- Designed mobile UI/UX and implemented PostHog analytics for 687 users, achieving 86.1% onboarding completion

Software Engineer (Part-time)

November 2025 – Present

PM Technologies LLC

El Paso, Texas

- Developed a Flask monitoring API that automated floor checks across 7 printers with 203 tests
- Applied hash-based deduplication to prevent print-record duplication caused by FAT32 truncation
- Ensured zero SQLite data corruption during power loss using write-ahead logging and atomic snapshot swaps
- Enabled zero-downtime schema migrations on a Raspberry Pi with idempotency guards and automated backups

Undergraduate Teaching Assistant

January 2025 – May 2025

The University of Texas at El Paso

El Paso, Texas

- Delivered 4 lab sessions and exam reviews with live coding demos for a 50-student lab
- Guided students in a 300-student lecture course through office hours and detailed coursework feedback

Machine Learning Researcher

May 2024 – July 2024

The Ohio State University AI-EDGE Institute

Columbus, Ohio

- Reduced Stable Diffusion 2 fine-tuning time by 55% (11h to 5h) using LoRA adapters
- Improved character renders by fine-tuning Stable Diffusion 2 across 6 curated Pokémon datasets

Projects

Sophia AI | Team of 3

January 2026 – April 2026

- Integrated live conversation audio, speaker diarization, and LLM parsing to generate Google and Apple Calendar events
 - Implemented primary-speaker identification for multi-party audio using enrolled voice profiles
 - Prevented duplicate SMS sends under concurrent workers with row-level locking and idempotency keys
- Python • FastAPI • PostgreSQL • Railway • Claude API

ReconAIssance | Team of 4 | 2nd Place, Hackathon | 600+ participants

October 2025 – March 2026

- Trained ArcFace and fine-tuned YOLOv8 with hard-negative mining for weapon and person-of-interest detection
 - Achieved per-frame risk scoring under 200ms by fusing model outputs in one async pass
- Python • FastAPI • OpenCV • YOLOv8 • InsightFace • React

Technical Skills

Languages: Python, Java, TypeScript, JavaScript, SQL

Frameworks: FastAPI, Flask, React, React Native, PyTorch

Data & Tools: PostgreSQL, SQLite, Supabase (SQL), Firebase (NoSQL), Docker, Git/GitHub, CI/CD, Linux, pytest